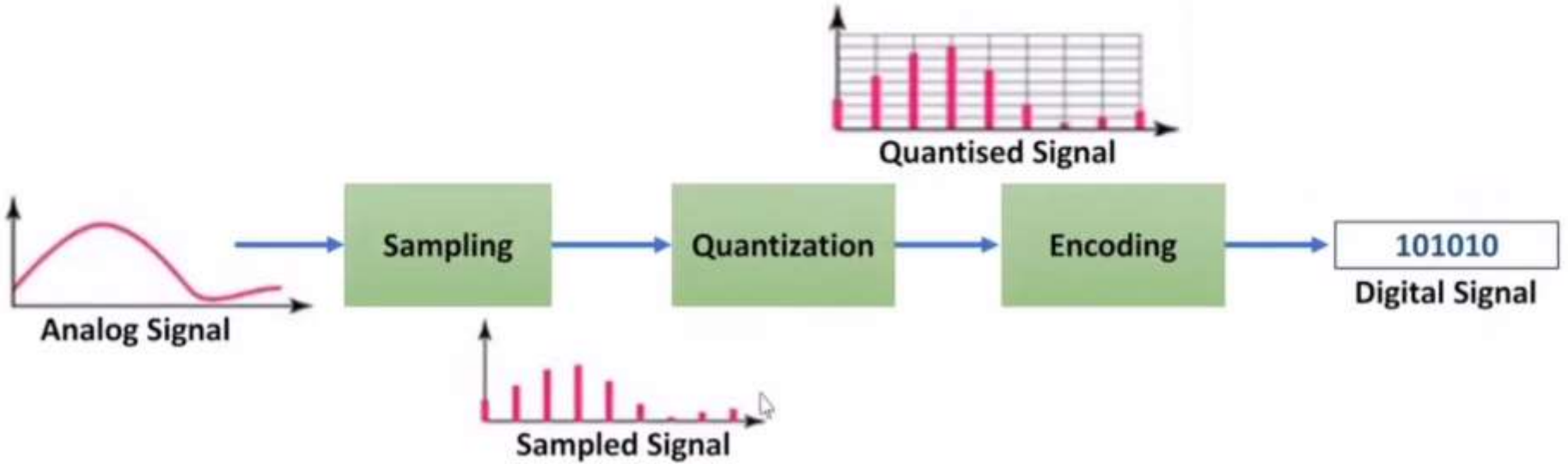
# Analog Signal to Digital Signal Conversion




- Sampling converts a Continuous-Time Signal into a Discrete-Time Signal.
- Quantisation converts a Discrete-time Signal into a Discrete-Time Discrete-Amplitude Signal.
- Encoding gives Digital values to each amplitude level with respect to samples.

# Basics of Sampling Theory

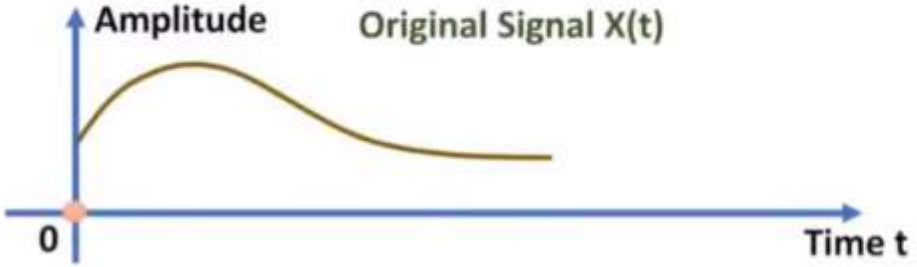
- Sampling is a Process to convert a Continuous-Time Signal into a Discrete-Time Signal.
- A sufficient number of samples must be taken to ensure that the original signal can be accurately reconstructed.
- The required sampling rate is determined by the maximum frequency present in the original signal and should be greater than the Nyquist rate.

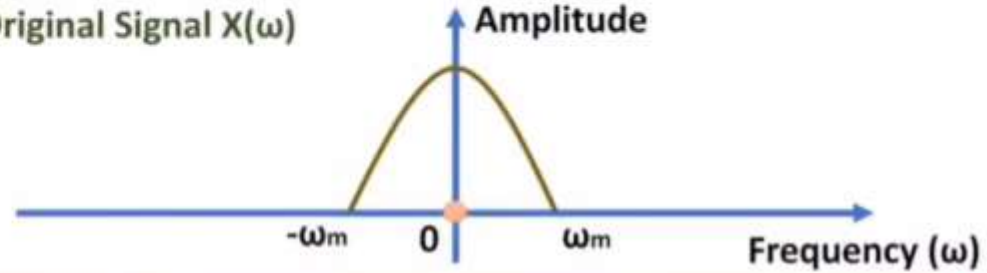
$$\Rightarrow F_s \geq 2F_m$$

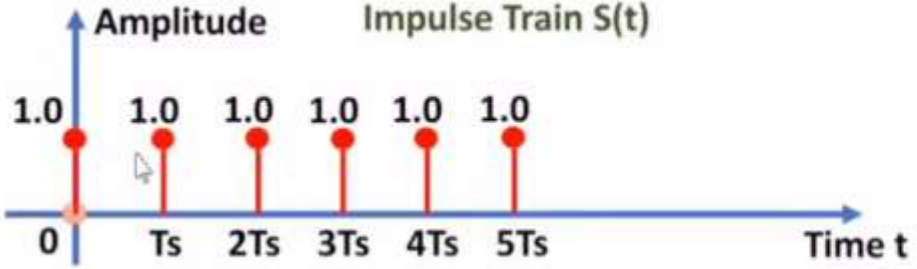
  - $F_s$ is Sampling Rate {Sampling Frequency}
  - $F_m$ is Maximum Frequency of Original Signal

- Types of Sampling
  - Ideal Sampling
  - Natural Sampling
  - Flat-Top Sampling

# Frequency Spectrum of Sampling



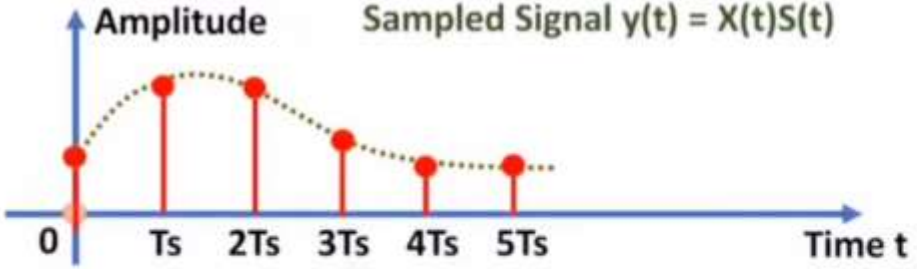



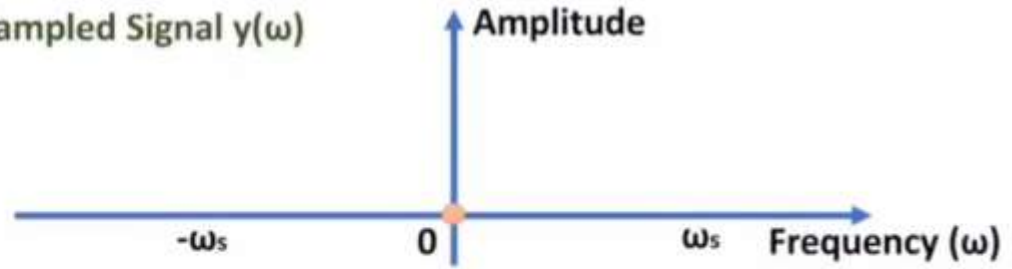




$$\Rightarrow s(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s) = \frac{1}{T_s}[1 + 2\cos(\omega_s t) + 2\cos(2\omega_s t) + \cdots]$$

$$\Rightarrow y(t) = \frac{1}{T_s}X(t)[1 + 2\cos(\omega_s t) + 2\cos(2\omega_s t) + \cdots]$$

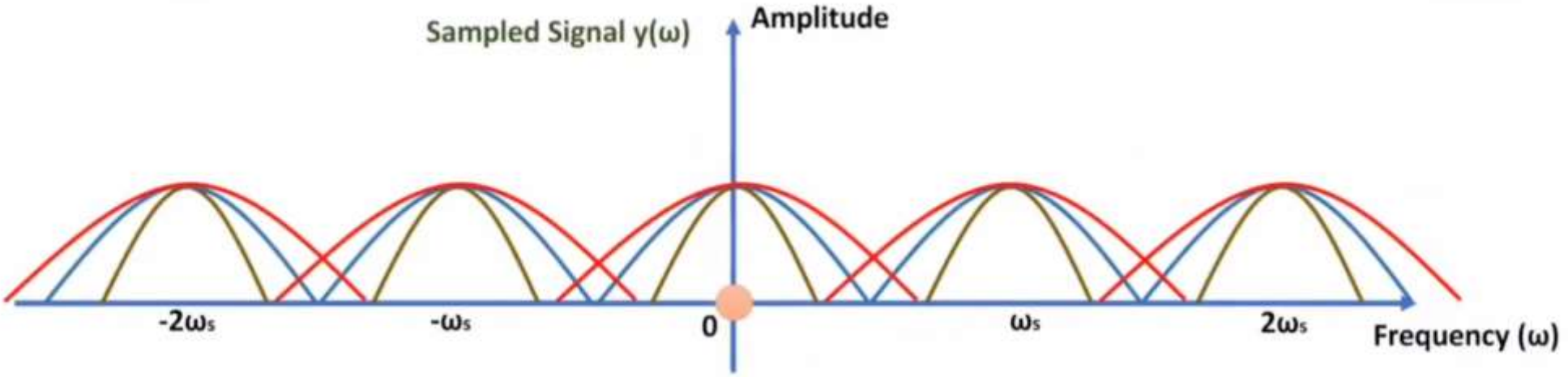
$$\Rightarrow y(\omega) = \frac{1}{T_s}[X(\omega) + X(\omega \pm \omega_s) + X(\omega \pm 2\omega_s) + \cdots]$$



# Aliasing Effect and Nyquist Rate

- Aliasing is a phenomenon that results from under-sampling and is also referred to as foldover error or under-sampling.
- Lets Consider Three Cases to understand Aliasing Effect:
  - Sampling Frequency is greater than Nyquist Rate (Fs > 2Fm)
  - Sampling Frequency is Equal to Nyquist Rate (Fs = 2Fm)
  - Sampling Frequency is less than Nyquist Rate (Fs < 2Fm)
- Prior to sampling, the original signal is filtered using a low-pass filter (LPF), commonly known as a pre-aliasing filter or band-limiting filter, to limit its frequency components.



## Key Steps to avoid Aliasing Effect

- Step-1: Sampling Rate $F_s$ Should be Greater than Nyquist Rate $2F_m$.
- Step-2: Use an Aliasing Filter before Sampling.
- Step-3: At the receiver, use a High Order (Sharp edge) Filter.
- Step-4: Keep Some Space in between successive bands of the sampled signal.

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